Liangjun Xing

Email ◇ Phone: +86 13337549946 ◇ GitHub ◇ LinkedIn

Education

Tsinghua University Beijing
Master of Automation, Supervised by Prof. Yebin Liu 09/2024- Present

Xi'an Jiaotong University Xi'an, Shaanxi
Bachelor of Automation 09/2020 - 07/2024

Leading Research Projects

Humanoid Loco-Manipulation WAM with Spatial Memory Internship in LingBot-VA Team, RobbyAnt
05/2026 - Present

- Proposed an embodiment-conditioned spatial memory: each visual patch is linked to its 3D viewing ray, and the robot's own intended movement decides which historical scene observations are recalled into current reasoning.
- Unified world prediction and action generation under a shared memory routing, keeping decisions correct even when task-relevant targets are out of the camera view.
- Designed progress-aware post-training that upweights actions measurably advancing the task, reducing early-rollout hesitation in long-horizon tasks under a single head-mounted camera.

Predictive Whole-Body Humanoid Teleoperation (PHT) Internship in LingBot-VA Team, RobbyAnt
2026 - Present

- Proposed PHT, a predictive whole-body teleoperation framework generating future reference horizons from irregular human observations, with conditional-flow retargeting and a residual MoE tracker (co-first author, under review).

Humanoid Force-Aware Whole-Body Controller and Teleop System Shanghai AI-Lab Internship · First author
09/2025 - 04/2026

- Developed a unified reinforcement learning control policy as well as teleoperation system for humanoid robot locomotion and manipulation without reliance on demonstration data, supporting torque-based hybrid force–position control across the whole body.
- Designed a base+residual hierarchical policy trained end-to-end without reference motion data, with a compact 20-dim action space (end-effector poses, hand open/close, base motion) validated on box carrying and pick-and-place in simulation and on real hardware.
- Built a 30 Hz multimodal data collection pipeline in C++ (RGB-D, LiDAR, proprioception, SLAM, high-level commands).
- Built DP and VLA command-generation frameworks on the collected data.

Humanoid Robots Playing Soccer Project of Internship in Unitree
05/2025-08/2025

- As a Unitree intern on the Tsinghua THMOS RoboCup team, developed decision-making, fall-recovery training, and a perception-driven 2-DoF head camera pipeline for real-time ball tracking.
- Won the 4th prize at the 2025 RoboCup in Brazil, marking the first appearance of Unitree's humanoid robot in robotic soccer competition.

Motion Tracking Methods for Dexterous-Hand Manipulation Master's Research Project
09/2024 - 06/2025

- Optimize the retargeting pipeline from MoCap data to dexterous hand interactions.
- Design reward functions for reinforcement learning trajectory trackers, especially articulated object manipulation.
- Address task expansion challenges using policy distillation modules.
- Adapted to motion generation methods and various kinds of Dexterous-Hand.

Publications

- **L. Xing***, K. Fan*, K. Li*, B. Wang, X. Niu, Y. Wang, et al., "FORGE: Force-Aware Multi-Granularity Control for Efficient Humanoid Loco-Manipulation," under review at CoRL 2026
- C. Zhu*, **L. Xing***, L. Li, Y. Liu, Y. Xu, "PHT: Predictive Whole-Body Humanoid Teleoperation," under review
- Y. Chai, L. Deng, R. Shao, J. Zhang, K. Lv, **L. Xing**, et al., "GAF: Gaussian Action Field as a 4D Representation for Dynamic World Modeling in Robotic Manipulation," ICRA 2026

- W. Yao, H. Ma, H. Zhang, **L. Xing**, Z. Yang, et al., "SynAgent: Generalizable Cooperative Humanoid Manipulation via Solo-to-Cooperative Agent Synergy," under review

Technical Skills

Python (PyTorch, NumPy)
Robot Simulation (Mujoco, Isaac Gym, Genesis)
Deep Reinforcement Learning (PPO, SAC, TD3)
Robot Operating System (ROS)
3D Vision (3DGS, NeRF, Pose Estimation, Digital Human)
C++ (Robotics Programming)
English: IELTS 7, GRE 320, CET-6 552

